Advanced Computer-Aided VLSI System Design
Homework 4: APR

TA: 張禾牧 r14943011@ntu.edu.tw **Due Tuesday, May 26, 13:59**

TA: 蔡岳峰 f12943014@ntu.edu.tw

Data Preparation

1. Get the 1142_hw4.tar.gz with following command

```
cp /share1/home/acvsd-ta09/1142_hw4.tar.gz .
```

2. Decompress 1142_hw4.tar.gz with following command

```
tar -zxvf 1142_hw4.tar.gz
```

Folder	File	Description
00_TESTBED	testfixture.v	Testbench for simulation w/o pad
00_TESTBED/ test_patterns	p*i*.dat	Input data (i = 0, 1, 2)
00_TESTBED/ test_patterns	p*i*_golden.dat	Output golden (i = 0, 1, 2)
00_TESTBED/axi	axi_ram.v	External memory
02_SYN/Netlist	top_syn.v	Synthesis netlist
02_SYN/Netlist	top_syn.sdc	Constraint for synthesis
02_SYN/Netlist	top-scan.def	Scan definition file
02_SYN/Netlist	top_syn.sdf	Timing info. from synthesis
03_GATE_PAD	03_run	Gate level simulation bash file
03_GATE_PAD	filelist.f	File list for gate level simulation
05_APR/lab_script	*.tcl	Scripts from APR Lab
06_POST	06_run	Post simulation bash file
06_POST	post_filelist.f	File list for post simulation

Introduction

In this homework, you should use Innovus to do P&R using the given design with scan chain. And try to add pad cell into netlist. You should modify the .sdc file and lab_scripts by yourself.

Pad Specifications

1. Pad cell name: PDCDG_V
2. You can refer to CHIP_syn.v in APR Lab to see how to add.
3. After adding pads, remember to modify the top module in testbench because the input/output ports are changed.
4. Make sure the modified Netlist pass every test pattern.

APR Specifications

1. Top module name: **top**
2. Follow mmmc.view from APR Lab to create the correct mmmc.view for this design
3. Add below command into top_syn.sdc to disable scan chain (port: test_se) for top_func.sdc and enable scan chain for top_scan.sdc

```
set_case_analysis...
```

   You can refer to CHIP_func.sdc and CHIP_scan.sdc in APR Lab to complete the command.

4. **For block design, Process related to IO Pad can be skip, but still need to create IO file**
5. **For block design, Process related to Power pad can be skip**
6. **For block design, Process related to Bump can be skip**
7. Process related to Early rail analysis can be skip
8. At least one power stripe in your design, but you should make sure IR drop < 8%
9. After finishing the APR, use below command to analyze the area (analyzeFloorplan destroys the design. Remember to save your design files first!!)

```
innovus #> analyzeFloorplan
```

10. Use below command to check the critical path

```
innovus #> report_timing -max_path 1
```

11. Submit one of the SDF files (*_ss0p72vm40c.sdf or *_ff0p88v125c.sdf) and rename it to top_pr.sdf
12. For IR drop analysis, use the .vcd file generated with *_ff0p88v125c.sdf and pattern2 (+define+p2)
13. **Review each script before execution, as some parts may require modification**
14. **If doing chip-level design, remember to modify "clk_ram" period to 2.5ns and "clk_top" period to 4.0ns**
15. **Allow the clock gating latch cell violation after reset complete in post-sim**

Design Description

1. Manually add pad cells into Netlist and rerun the gate level simulation
2. Perform place & route using Innovus.
3. Run simulation after APR

Submission

1. Create a folder named **studentID_hw4**, and put all below files into the folder

```
r14943011_hw4
├── 00_TESTBED
│   └── testfixture_pad.v    (Testbench for modified netlist with pad cells)
├── 03_GATE_PAD
│   └── top_syn_pad.v        (modified netlist with pad cells)
└── 05_APR
    ├── final.enc            (APR design file)
    ├── final.enc.dat/       (APR design file data)
    ├── mmmc.view
    ├── top.gds
    ├── top_func.sdc
    ├── top_pr.sdf
    ├── top_pr.v
    └── top_scan.sdc
```

Note: Use **lower case** for the letter in your student ID. (Ex. r14943011_hw4)

2. Compress the folder **studentID_hw4** in a **tar.gz file** named **studentID_hw4.tar.gz**

```
tar -zcvf studentID_hw4.tar.gz studentID_hw4
```

- TA will only check the last version of your homework.
- Place the tar.gz file at the home directory (~) of your ADFP account
- Note: Use **lower case** for the letter in your student ID. (Ex. r14943011_hw4)

3. Submit the report named **studentID_report.pdf** in NTU COOL

- Note: Use **lower case** for the letter in your student ID. (Ex. r14943011_report.pdf)

Grading Policy

1. TA will run your code as the commands in 03_run and 06_run. Make sure to run these commands with no error message for each test pattern.
2. APR report: **50%**
3. Correctness of simulation after adding pads: **10%**
4. Correctness of mmmc.view setting: **5%**
5. Correctness of top_func.sdc and top_scan.sdc setting: **5%**
6. Correctness of simulation after APR: **10%**

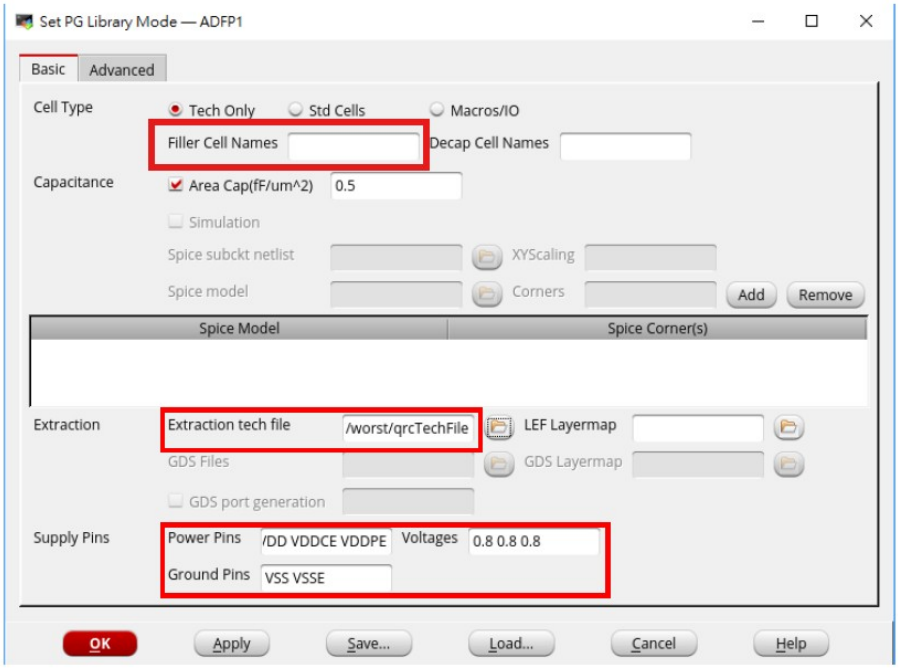
7. For Chip-level design with IO pads and bumps: **20%**
8. 5 points minus for any incorrect naming or submission format
9. No late submission, or 0 point for this homework

Appendix – Power Analysis

1. Generate PG Library (Since HW4 doesn't do the Early Rail Analysis, this is also the first step for Power Analysis)

2. Open ***Power → Rail Analysis → Set PG Library Mode***

   Remember to add Filler Cell Names if you have already added core fillers



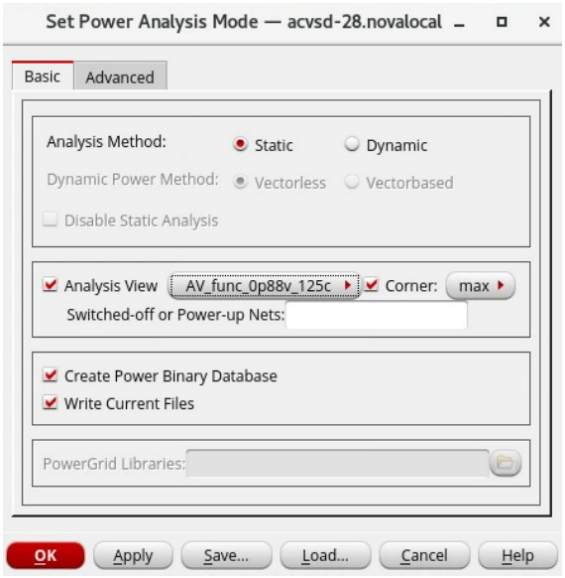
3. Open ***Power → Rail Analysis → Generate PG Library***



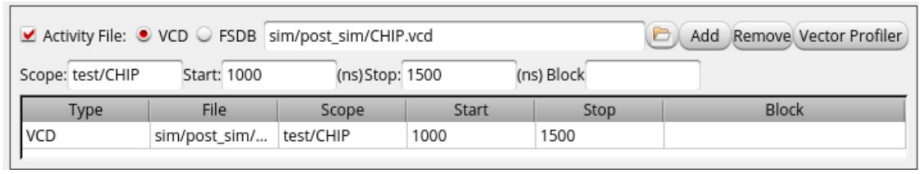
   Click OK

4. **Run post layout simulation** to get the vcd file
5. Follow the remaining steps in Lab5 power analysis part

6. Open ***Power → Power Analysis → Setup…***

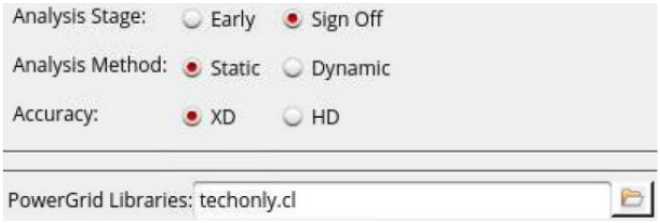


7. Open ***Power → Power Analysis → Run…***

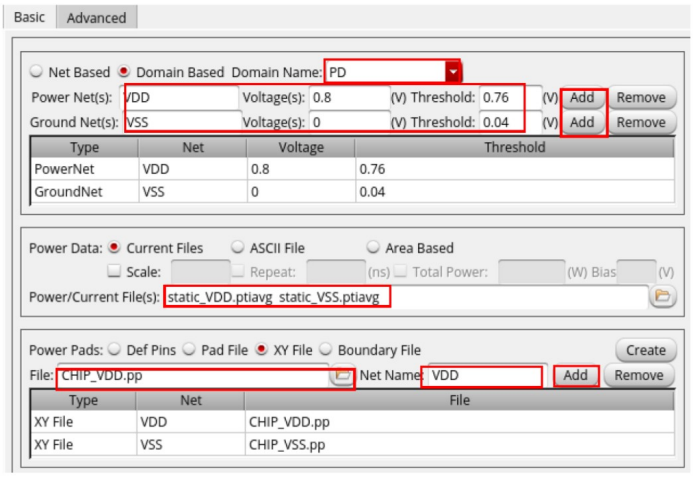


Remember to change the settings

8. Open ***Power → Rail Analysis → Setup Rail Analysis…***

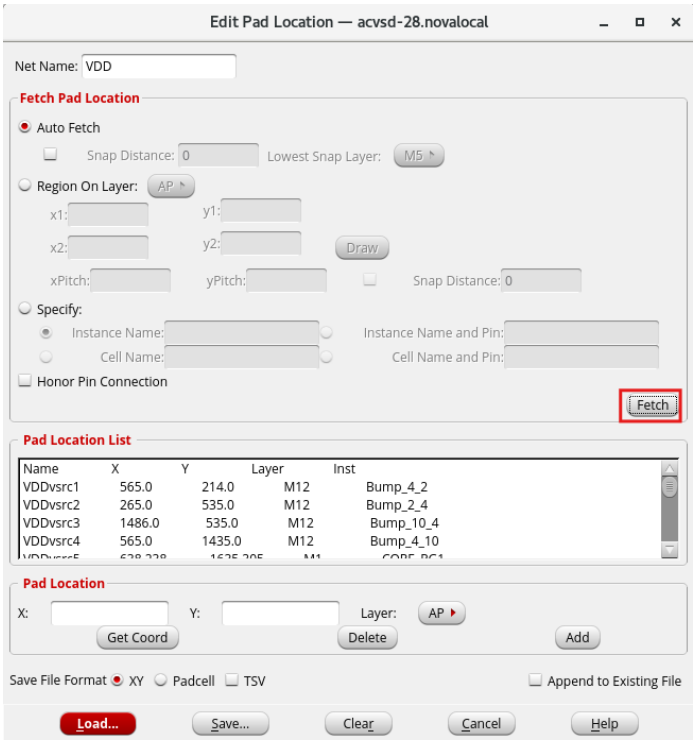


9. Open ***Power → Rail Analysis → Run Rail Analysis…***



10. Gernerate *.pp files by create button

11. In XY File creating Gui, press Fetch button, and then save the files as *_VDD.pp and *_VSS.pp



12. Find the file **ir_linear.gif** under the path …/PD_25C_avg_x/VDD and VSS to see if the IR drop meets the requirement